

UNIVERSITY OF THE PHILIPPINES
Tacloban College
Division of Natural Sciences and Mathematics
Bachelor of Science in Computer Science

CMSC 12 — Fundamentals of Programming 2
Second Semester AY 2025-2026

MACHINE PROBLEM

Final Report

A. Project Title

Memoriam: Aspect of Fate

B. Brief Description

Memoriam: Aspect of Fate is a top down dungeon-crawler game inspired by the combat progression of Cookie Run Adventure mini-game from Cookie Run Kingdom while adapting the exploration style often associated with Legends of Zelda. Although the game takes inspiration from these games, it introduces its own mechanisms and systems that emphasize replayability, strategy, and player choice.

The game follows a stage-based progression system where players advance through five combat-focused levels before facing a final boss, with an additional infinite survival mode available separately.

The game operates through a custom build system featuring scene management, entity handling, collision detection, player progression systems, and save-state persistence. The game operates through a custom-built engine featuring scene management, entity handling, collision detection, player progression systems, and save-state persistence. Gameplay occurs in real time at 30-60 frames per second, with movement controlled through keyboard input and combat directed using mouse-based aiming and projectile shooting.

Players control a character capable of moving freely within each level while fighting enemy variants with distinct behaviors and attack patterns. Enemies detect, pursue, and attack the player using melee or projectile-based combat, while bosses feature enhanced statistics and multi-phase mechanics. Combat interactions are managed through a collision system using rectangular colliders to detect impacts between players, enemies, projectiles, and environmental barriers.

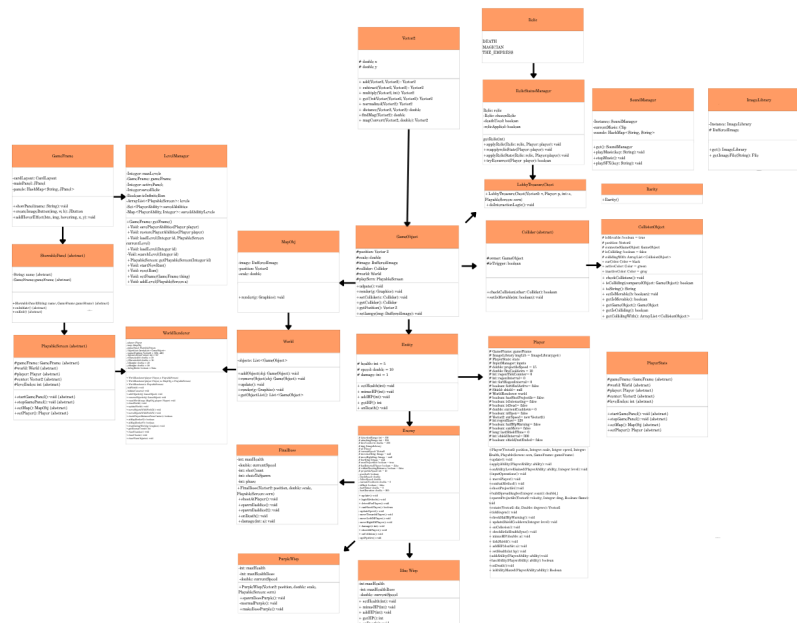
UNIVERSITY OF THE PHILIPPINES
Tacloban College
Division of Natural Sciences and Mathematics
Bachelor of Science in Computer Science

Progression is driven by a tarot card-based ability system. After completing a level, players are presented with randomized tarot cards which grant different effects such as health regeneration, multi-shot projectiles, shields, flame attacks, and movement enhancements. These abilities modify player statistics dynamically through additive and percentage based modifiers. In addition, relics, tarot cards from the Major Arcana, obtained in the lobby, provide powerful permanent effects that persist throughout the run.

The game includes a persistent save system that stores player progress, abilities, relics, health, and level progression. Between levels, the player's abilities and statistics are serialized and restored to maintain continuity across gameplay sessions.

An alternate infinite mode provides endless wave-based combat with progressively randomized enemies, focusing purely on survival, gameplay without progression or save based mechanics.

C. UML Class Diagram



UNIVERSITY OF THE PHILIPPINES
Tacloban College
Division of Natural Sciences and Mathematics
Bachelor of Science in Computer Science

D. Reflection

A short reflection on how you addressed thread synchronization and which design patterns (e.g., Singleton, Observer, or Factory) were utilized.

At the start of the project, a key system that helped us in the long run was figuring out the system for switching panels. They all stem from our custom made ShowablePanel class, this class had 2 key methods, OnInitiate() and OnExit(), these methods were both used by a forLoop that can be found in the GameFrame class, these methods essentially acted as flags for our panels to perform methods and start threads that are related to the inner workings of the class.

For Singletons, they were used for managers like ImageLibrary and SoundManager in order to ensure only one global instance that the entire project can pull from. The Factory and Observer Ydesign patterns were implemented for the more complex actions we wanted to be more flexible on such as the creation of some enemy subtypes (miniBosses), SpecialEffects class, and the Collision system of our game.